

Jianqi Ruan

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EDUCATION

- Ph.D., Mechanical Engineering**, Purdue University, West Lafayette, IN Expected Dec. 2023
Specialization: System, Control, and Reinforcement Learning. 3.89 GPA
Advisor: Dr. Neera Jain
Thesis: *Reinforcement Learning-Based Human Operator Decision Support Agent for Highly Transient Industrial Processes*
- B.S., Mechanical Engineering**, Pennsylvania State University, State College, PA May 2019
Honors: Annual Dean's List of 4 consecutive academic years 3.95 GPA

RESEARCH INTERESTS

Deep Learning; Reinforcement Learning; Unsupervised Clustering; Constrained Optimization; Optimal Control; Dynamic Modeling

RESEARCH EXPERIENCE

- Graduate Research Assistant**, Purdue University May 2019 - Present
- Conducted human subject studies to understand decision making processes of human operators in an industrial process
 - Developed and applied similarity metric and unsupervised clustering methods to categorize decision making processes
 - Analyzed the performance (with respect to control objectives & time efficiency) of different types of decision making behaviors
 - Designed, trained, and evaluated reinforcement learning agents to provide suggestions of multiple control setpoints real time adjustments
 - Designed, constructed, and verified a training-free data-driven similarity-based plant simulator for the industrial process
- Undergraduate Research Assistant**, Pennsylvania State University Aug. 2018 – May 2019
- Designed a 3D printable roach robot
 - Analyzed the stress concentration of the roach robot leg portion with FEA
 - Optimized the joint's shape and size to extend its fatigue life

PUBLICATIONS

Peer Reviewed Journal Articles

- **J. Ruan**, B. Nooning, I. Parkes, W. Blejde, G. Chiu, and N. Jain, "Human Operator Decision Support for Highly Transient Industrial Processes: A Reinforcement Learning Approach," *ACM Transactions on Human-Robot Interaction*. (Under Review)

Peer Reviewed Conference Papers

- **J. Ruan**, M. Ponder, I. Parkes, W. Blejde, G. Chiu, and N. Jain, "Applying a Deep Q-Network for Human Operator Behavioral Modeling and Decision Support in a Twin-Roll Casting Process," in *2022 American Control Conference (ACC)*, Atlanta, GA, 2022, pp. 689-696

- **J. Ruan**, B. Nooning, I. Parkes, W. Blejde, G. Chiu, and N. Jain, “A Training-Free Data-Driven Method for Input-Output Modeling of Complex Processes,” in *2022 Modeling, Estimation and Control Conference*, Jersey City, NJ, 2022. (Presented)

PROFESSIONAL ACTIVITIES

Conference Manuscript Reviewer

- Modeling, Estimation, and Control Conference 2022
- IFAC World Congress 2023